$$\bar{x} = A + \frac{\sum_{i=1}^{n} f_i d_i}{N} \times C$$

$$= 28 + \frac{-25}{77} \times 8 \quad = 25.40$$

### Merits

- It is easy to compute and has a unique value.
- It is based on all the observations.
- It is well defined.
- It is least affected by sampling fluctuations.
- It can be used for further statistical analysis.

### Limitations

- The mean is unduly affected by the extreme items (outliers).
- It cannot be determined for the qualitative data such as beauty, honesty etc.
- It cannot be located by observations on the graphic method.

**When to use?**

Arithmetic mean is a best representative of the data if the data set is homogeneous. On the other hand if the data set is heterogeneous the result may be misleading and may not represent the data.

## Weighted Arithmetic Mean

The arithmetic mean, as discussed earlier, gives equal importance (or weights) to each observation in the data set. However, there are situations in which values of individual observations in the data set are not of equal importance. Under these circumstances, we may attach, a weight, as an indicator of their importance to each observation value.

**Definition**

Let $x_1, x_2, ...., x_n$ be the set of n values having weights $w_1, w_2, ...., w_n$ respectively, then the weighted mean is,

$$\overline{x_w} = \frac{w_1 x_1 + w_2 x_2 + ... + w_n x_n}{w_1 + w_2 + ... + w_n} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$



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